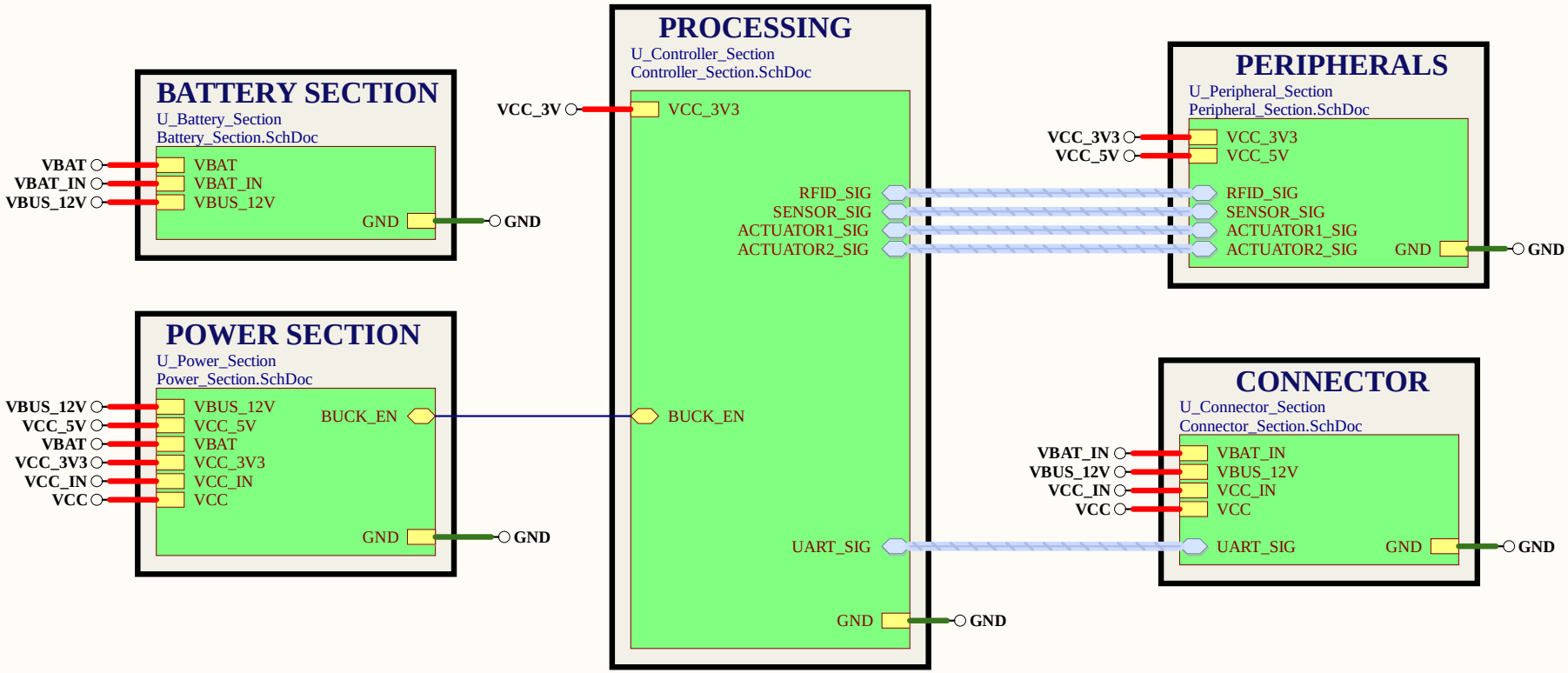
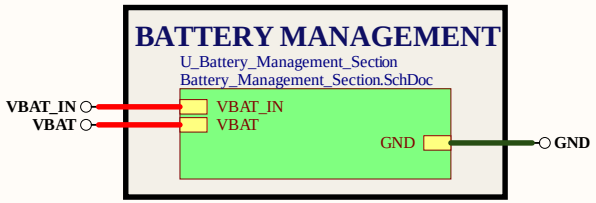
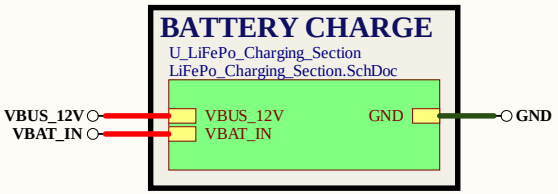
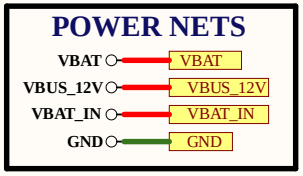
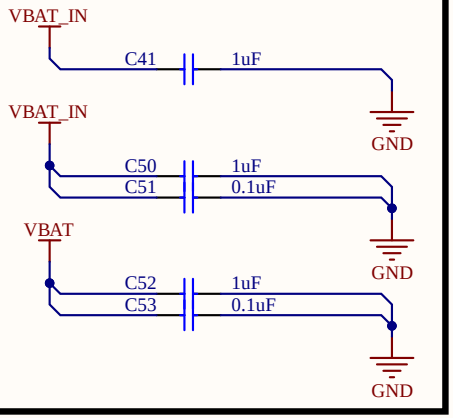

BATTERY SECTION
U_Battery_Section
Battery_Section.SchDoc
VBAT
VBAT_IN
VBUS_12V
GND
VBAT
VBAT_IN
VBUS_12V
GND
POWER SECTION
U_Power_Section
Power_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
BUCK_EN
GND
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
BUCK_EN
GND
PROCESSING
U_Controller_Section
Controller_Section.SchDoc
VCC_3V3
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
BUCK_EN
UART_SIG
GND
VCC_3V3
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
UART_SIG
GND
PERIPHERALS
U_Peripheral_Section
Peripheral_Section.SchDoc
VCC_3V3
VCC_5V
RFID_SIG
SENSOR_SIG
ACTUATOR1_SIG
ACTUATOR2_SIG
GND
VCC_3V3
VCC_5V
GND
CONNECTOR
U_Connector_Section
Connector_Section.SchDoc
VBAT_IN
VBUS_12V
VCC_IN
VCC
UART_SIG
GND
VBAT_IN
VBUS_12V
VCC_IN
VCC
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND


BATTERY MANAGEMENT
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT_IN
VBAT
VBAT_IN
VBAT
GND
GND

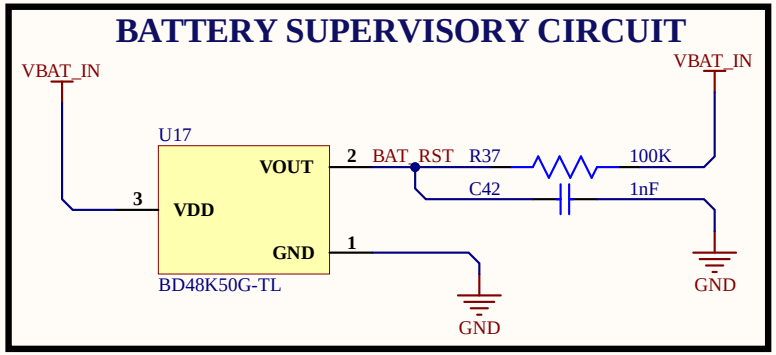

BATTERY CHARGE
U_LiFePo_Charging_Section
LiFePo_Charging_Section.SchDoc
VBUS_12V
VBAT_IN
VBUS_12V
VBAT_IN
GND
GND


POWER NETS
VBAT
VBUS_12V
VBAT_IN
GND
VBAT
VBUS_12V
VBAT_IN
GND

DECOUPLING CAPACITOR

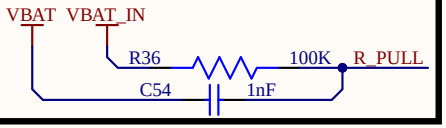


BATTERY SUPERVISORY CIRCUIT

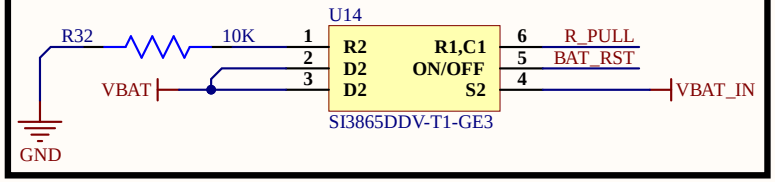


Detection voltage: 2.3V to 6.0V (Typ.)
Voltage threshold: 5V

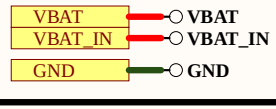
PULL UP RESISTOR



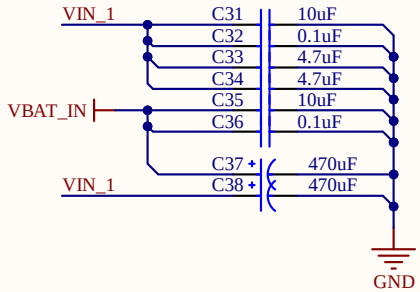
POWER SWITCH IC



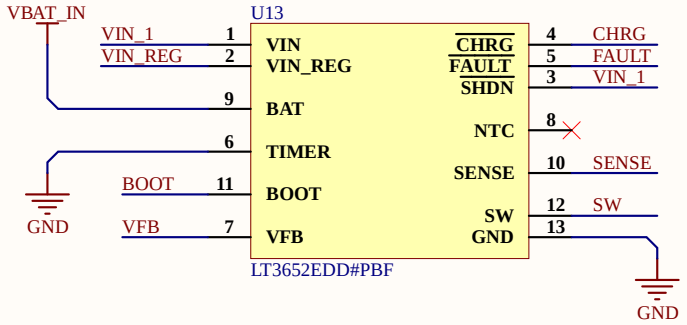
POWER NETS



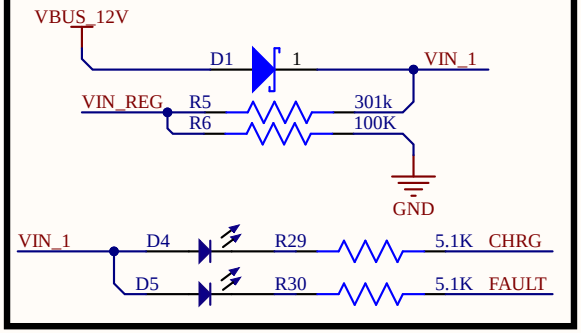
DECOUPLING CAPACITORS



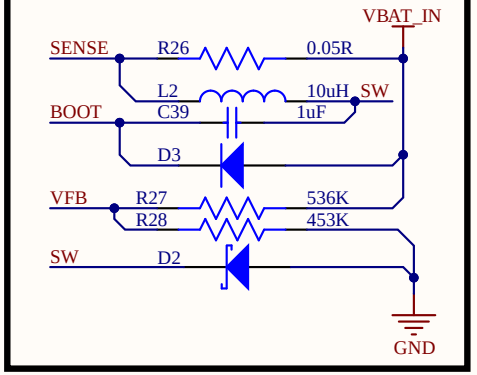
BATTERY CHARGING CIRCUIT



SUPPORTING CIRCUITRY

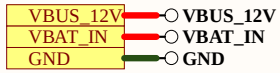


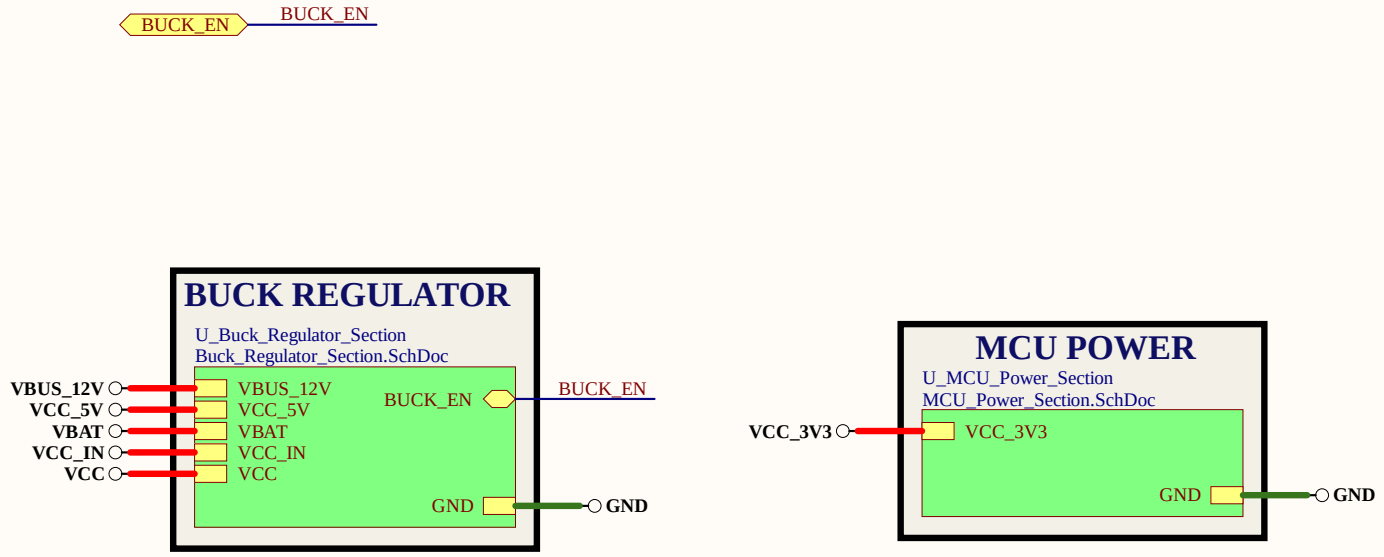
BATTERY INPUT CIRCUIT



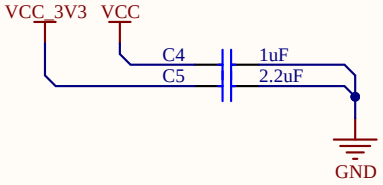
Battery Chemistry: 2 Cell LiFeOP4
Battery Voltage: (2 × 3.6V) = 7.2V
Charge current: 2A

POWER NETS

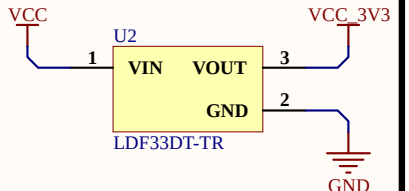



BUCK_EN
BUCK_EN
BUCK REGULATOR
U_Buck_Regulator_Section
Buck_Regulator_Section.SchDoc
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
VBUS_12V
VCC_5V
VBAT
VCC_IN
VCC
BUCK_EN
BUCK_EN
GND
GND
MCU POWER
U_MCU_Power_Section
MCU_Power_Section.SchDoc
VCC_3V3
VCC_3V3
GND
GND

DECOUPLING CAPACITOR

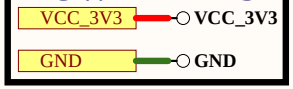


3V3 REGULATOR

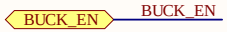


Input Voltage:
Output Current: 1A
Output Voltage: Fixed 3.3V
Voltage Drop: 500 mV max. at 1 A load

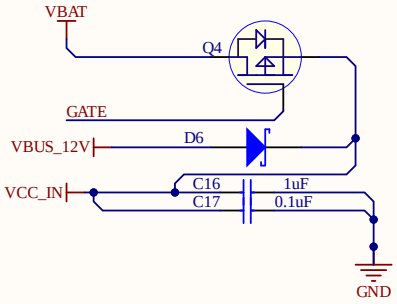
POWER NETS



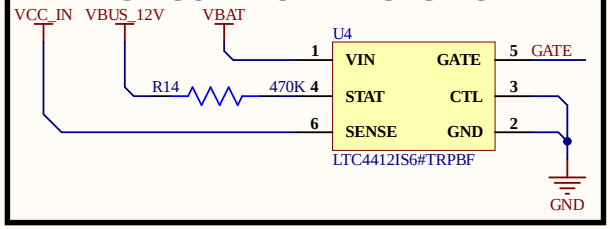
BUCK ENABLE



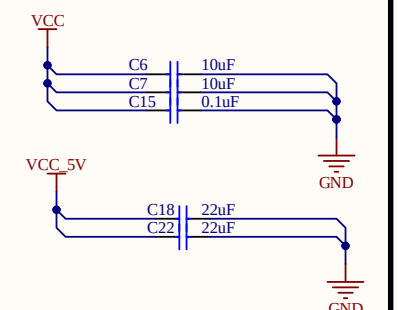
SUPPORTING CIRCUITRY



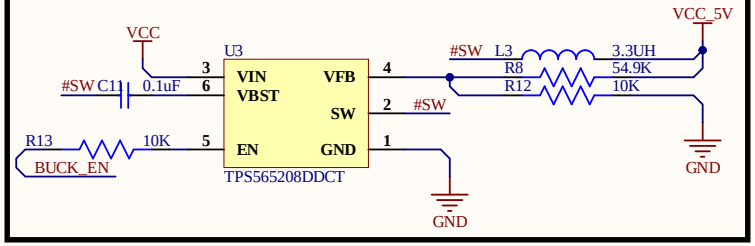
OR CONTROLLER SECTION



DECOUPLING CAPACITOR



BUCK SWITCHING REGULATOR CIRCUIT

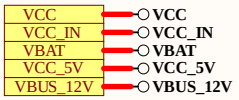


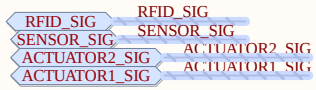
- Input Voltage: switches between 12V USB and VBAT
- Output Voltage: 5V
- Output Current: 5A Maximum.

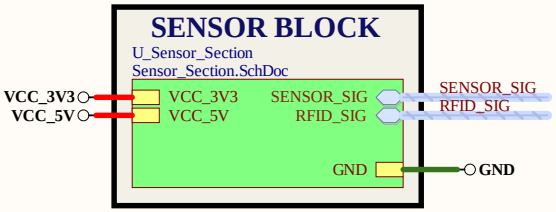
Output Voltage Calculation:

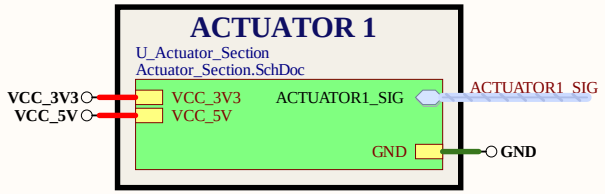
Formula: V =0.760 (1+R1/R2) (pg:13)
R1= 54.9K, R2=10K, L=3.3uH (pg:14)
VOUT= 4.93V
UVLO threshold: 3.6V (pg:5)

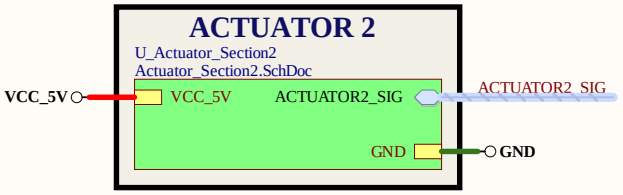
POWER NETS

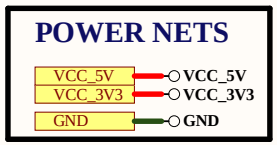


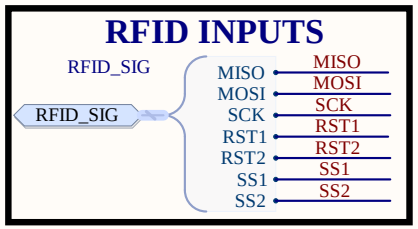

RFID_SIG
SENSOR_SIG
ACTUATOR2_SIG
ACTUATOR1_SIG
RFID_SIG
SENSOR_SIG
ACTUATOR2_SIG
ACTUATOR1_SIG

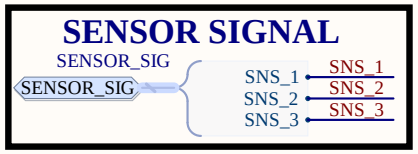

SENSOR BLOCK
U_Sensor_Section
Sensor_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
SENSOR_SIG
RFID_SIG
GND
SENSOR_SIG
RFID_SIG
GND


ACTUATOR 1
U_Actuator_Section
Actuator_Section.SchDoc
VCC_3V3
VCC_5V
VCC_3V3
VCC_5V
ACTUATOR1_SIG
GND
ACTUATOR1_SIG
GND

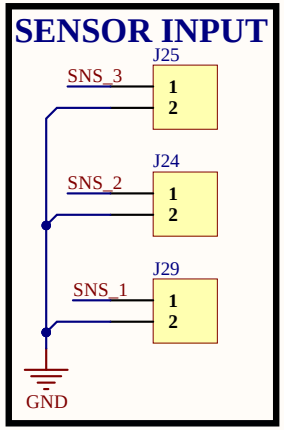

ACTUATOR 2
U_Actuator_Section2
Actuator_Section2.SchDoc
VCC_5V
VCC_5V
ACTUATOR2_SIG
GND
ACTUATOR2_SIG
GND

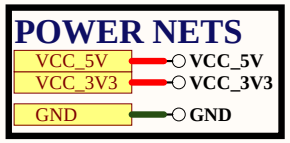

POWER NETS
VCC_5V
VCC_3V3
GND
VCC_5V
VCC_3V3
GND

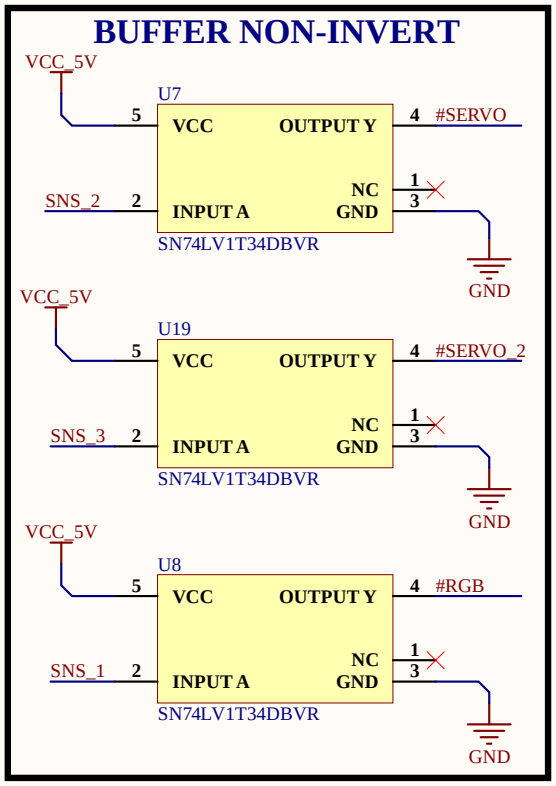


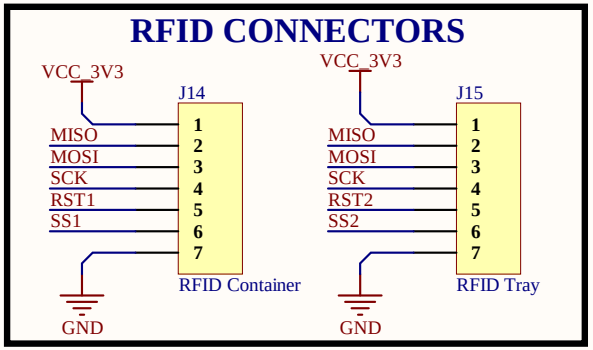




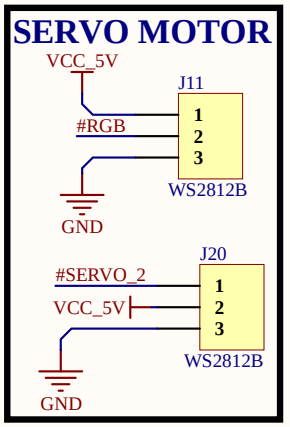


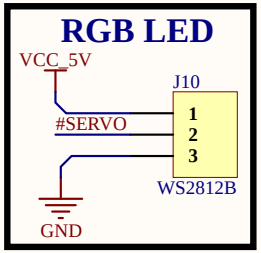


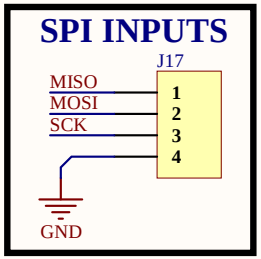




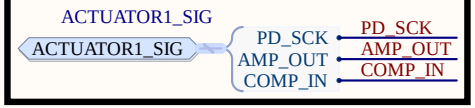
SNS signals are test signals for motor an d RGB LED.



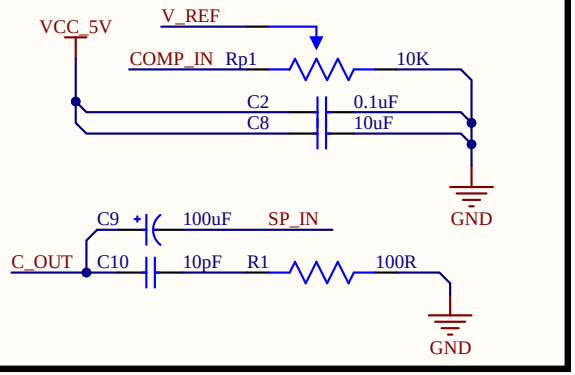




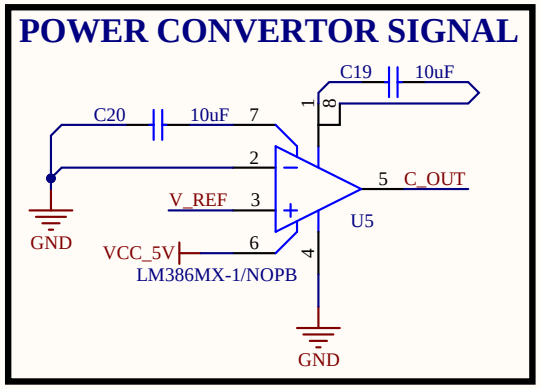
ACTUATOR1 SIGNAL



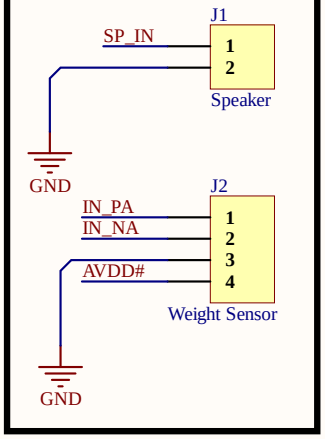
COMPARATOR SUPPORTING CIRCUIT



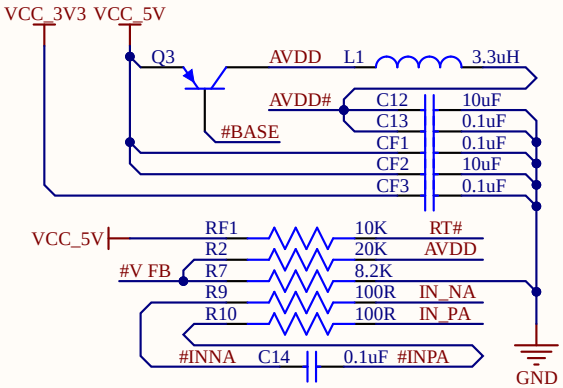
POWER CONVERTOR SIGNAL



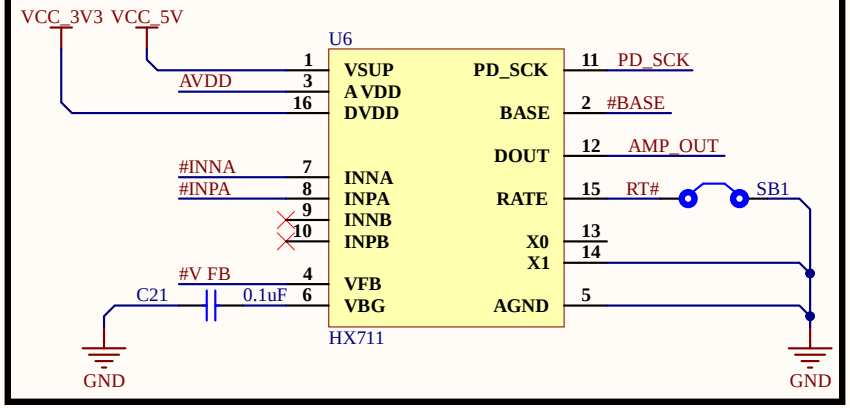
CONNECTORS



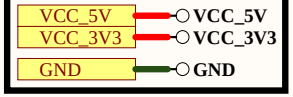
AMPLIFIER SUPPORTING CIRCUIT



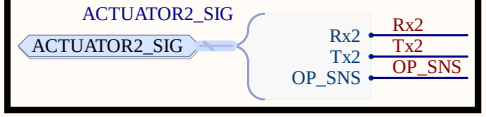
LOAD CELL AMPLIFIER



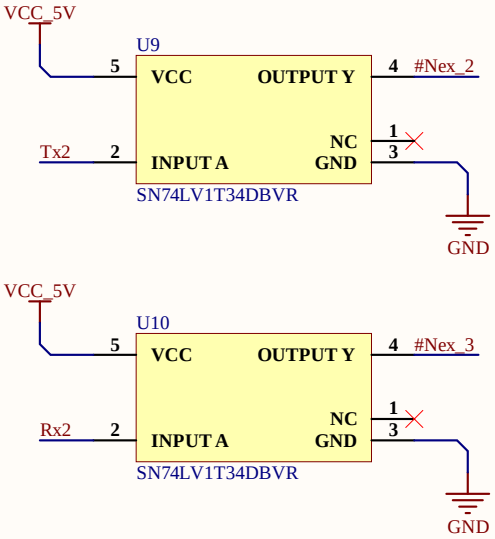
POWER NETS



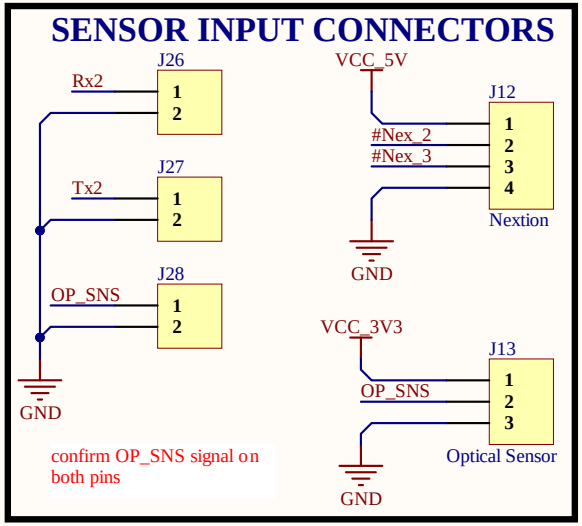
ACTUATOR2 SIGNAL



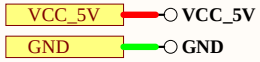
BUFFER CIRCUIT

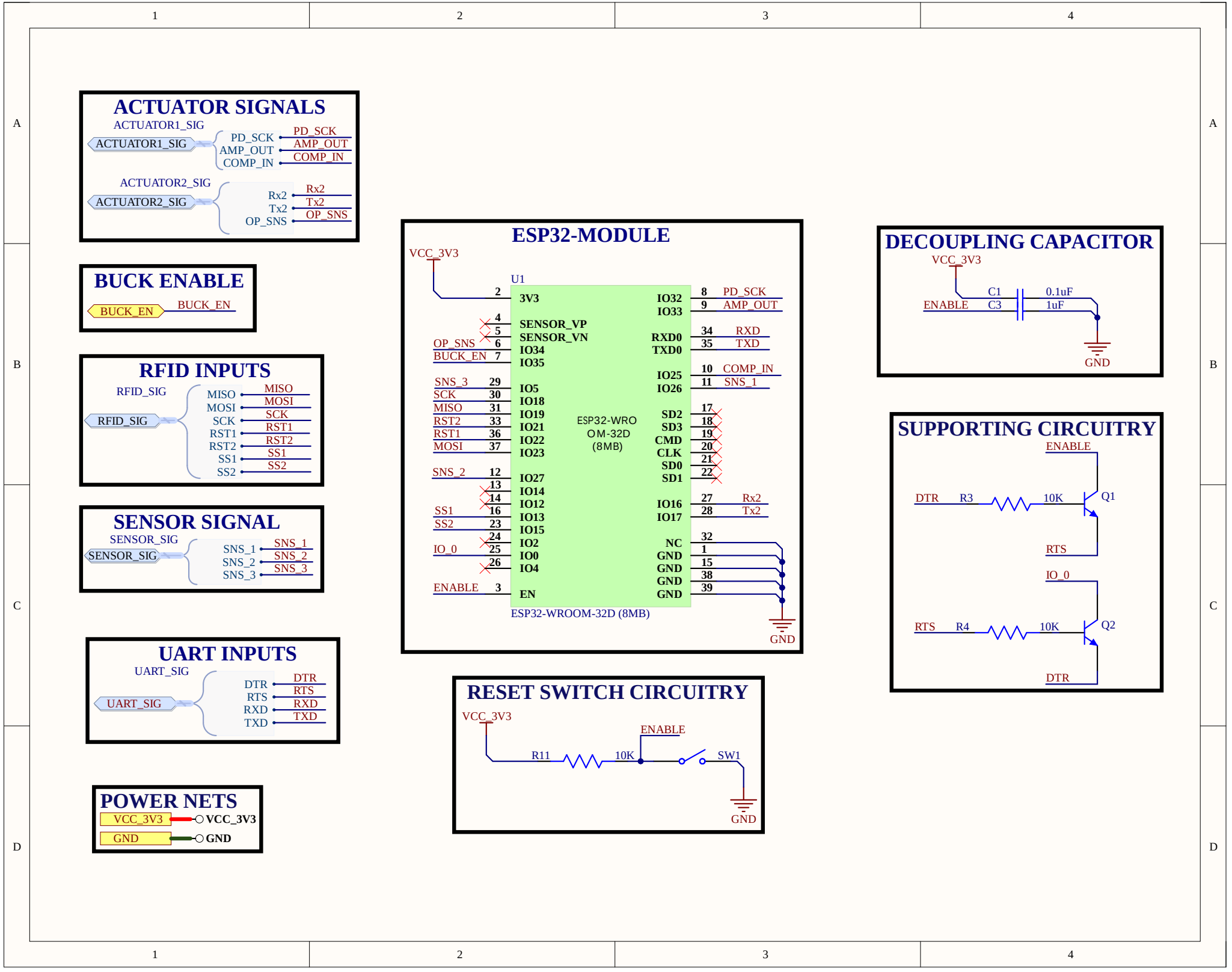


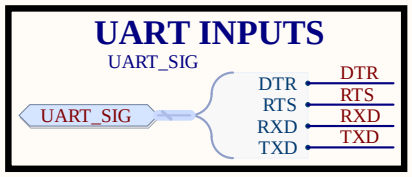
SENSOR INPUT CONNECTORS

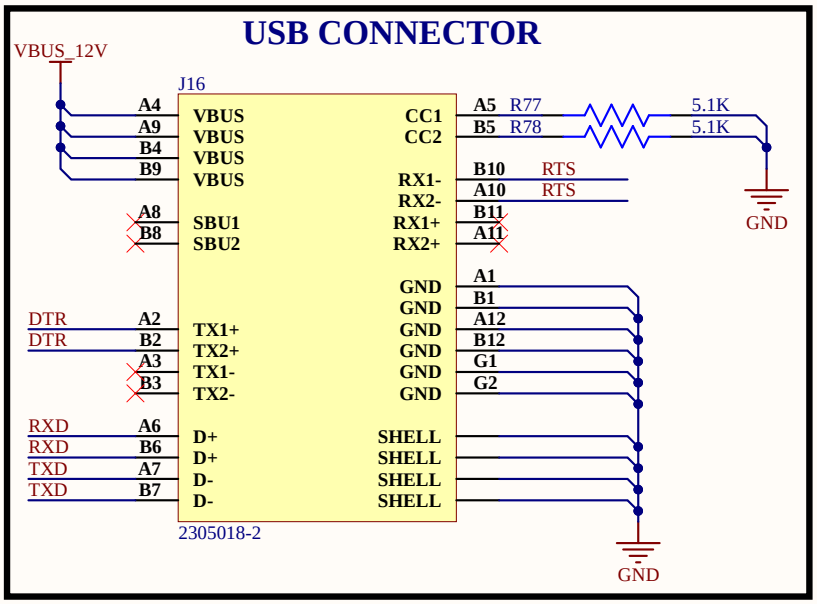


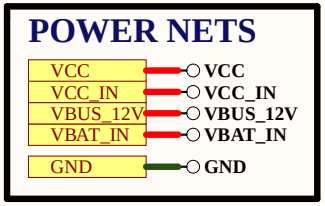
POWER NETS

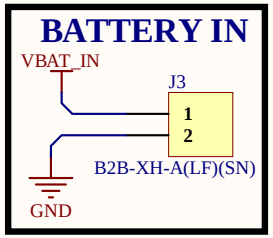


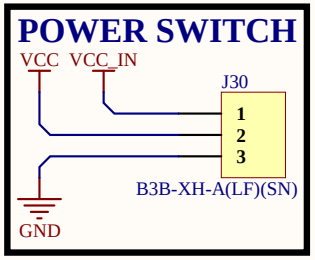

ACTUATOR SIGNALS
ACTUATOR1_SIG
ACTUATOR1_SIG
PD_SCK
AMP_OUT
COMP_IN
PD_SCK
AMP_OUT
COMP_IN
ACTUATOR2_SIG
ACTUATOR2_SIG
Rx2
Tx2
OP_SNS
Rx2
Tx2
OP_SNS
BUCK ENABLE
BUCK_EN
BUCK_EN
RFID INPUTS
RFID_SIG
RFID_SIG
MISO
MOSI
SCK
RST1
RST2
SS1
SS2
MISO
MOSI
SCK
RST1
RST2
SS1
SS2
SENSOR SIGNAL
SENSOR_SIG
SENSOR_SIG
SNS_1
SNS_2
SNS_3
SNS_1
SNS_2
SNS_3
UART INPUTS
UART_SIG
UART_SIG
DTR
RTS
RXD
TXD
DTR
RTS
RXD
TXD
POWER NETS
VCC_3V3
VCC_3V3
GND
GND
ESP32-MODULE
VCC_3V3
U1
2 3V3
4
5 SENSOR_VP
6 SENSOR_VN
OP_SNS 6 IO34
BUCK_EN 7 IO35
SNS_3 29 IO5
SCK 30 IO18
MISO 31 IO19
RST2 33 IO21
RST1 36 IO22
MOSI 37 IO23
SNS_2 12 IO27
13 IO14
14 IO12
SS1 16 IO13
SS2 23 IO15
24 IO2
IO_0 25 IO0
26 IO4
ENABLE 3 EN
ESP32-WRO
OM-32D
(8MB)
IO32 8 PD_SCK
IO33 9 AMP_OUT
RXD0 34 RXD
TXD0 35 TXD
IO25 10 COMP_IN
IO26 11 SNS_1
SD2 17
SD3 18
CMD 19
CLK 20
SD0 21
SD1 22
IO16 27 Rx2
IO17 28 Tx2
NC 32
GND 1
GND 15
GND 38
GND 39
ESP32-WROOM-32D (8MB)
GND
DECOUPLING CAPACITOR
VCC_3V3
ENABLE
C1
C3
0.1uF
1uF
GND
SUPPORTING CIRCUITRY
ENABLE
DTR
R3
10K
Q1
RTS
IO_0
RTS
R4
10K
Q2
DTR
RESET SWITCH CIRCUITRY
VCC_3V3
ENABLE
R11
10K
SW1
GND


UART INPUTS
UART_SIG
UART_SIG
DTR
RTS
RXD
TXD
DTR
RTS
RXD
TXD


USB CONNECTOR
VBUS_12V
J16
A4
A9
B4
B9
VBUS
VBUS
VBUS
VBUS
A8
B8
SBU1
SBU2
DTR
DTR
A2
B2
A3
B3
TX1+
TX2+
TX1-
TX2-
RXD
RXD
TXD
TXD
A6
B6
A7
B7
D+
D+
D-
D-
2305018-2
CC1
CC2
A5
B5
R77
R78
5.1K
5.1K
GND
RX1-
RX2-
RX1+
RX2+
B10
A10
B11
A11
RTS
RTS
GND
GND
GND
GND
GND
GND
A1
B1
A12
B12
G1
G2
SHELL
SHELL
SHELL
SHELL
GND


POWER NETS
VCC
VCC_IN
VBUS_12V
VBAT_IN
GND
VCC
VCC_IN
VBUS_12V
VBAT_IN
GND


BATTERY IN
VBAT_IN
J3
1
2
B2B-XH-A(LF)(SN)
GND


POWER SWITCH
VCC
VCC_IN
J30
1
2
3
B3B-XH-A(LF)(SN)
GND